



UNIVERSITY OF GHANA
COLLEGE OF BASIC AND APPLIED SCIENCES

**FATIGUE DETECTION FOR HAZMAT DRIVERS IN
WEST AFRICA USING EYE-STATE AND FACIAL
LANDMARK ANALYSIS**

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Declaration

STUDENT

I hereby declare that this thesis is the result of my own original research and that no part of it has been presented for another degree in this University or elsewhere. Every claim that rests on another investigator's measurement is cited in APA 7th edition. I have not inserted placeholder papers. The companion software is public at the repository named in Appendix A.

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Abstract

Context:

Petroleum tankers on West African trunk roads, including the Tema–Kumasi and Lagos–Ibadan corridors, move bulk fuel in cabins that are dark at night, optically dirty in the Harmattan, and staffed by drivers whose duty length is set by trip pay rather than by a laboratory protocol. The World Health Organization (2023a) put annual road traffic deaths at 1.19 million in 2021. The Africa Status Report estimated 259,601 deaths on the continent that year (SSATP, 2025). Fatigue is one mechanism that can take a loaded tanker out of lane. It is not the only mechanism. A camera does not replace brake inspections or hours-of-service rules. What a camera can attempt is to measure eyelid geometry without electrodes that a bulk-road driver will not wear (Sahayadhas et al., 2012; Sikander & Anwar, 2019).

Aim:

This study specifies, implements, and bounds a non-intrusive eye-state and facial-landmark detector written for that cabin rather than copied from a passenger-car notebook. The algorithm is named WA-PERCLOS-HYS.

Method:

MediaPipe Face Mesh supplies dense landmarks on a handset (Kartynnik et al., 2019; Grishchenko et al., 2020). From those points the system computes an exponentially smoothed Eye Aspect Ratio after Soukupová and Čech (2016), a mouth ratio, rolling PERCLOS in the P80 tradition (Wierwille et al., 1994; Dinges & Grace, 1998), the share of long-closure blinks (Caffier et al., 2003; Schleicher et al., 2008), and a coarse head-pitch cue. Closure is decided with a two-threshold hysteresis gate. The close threshold is 68 percent of the driver’s own open-eye EAR, collected in an eight-second depot calibration and scaled when cabin luma indicates a night haul. Five evidence channels are fused in the spirit of Ji et al. (2004) and Bergasa et al. (2006). Eyes shut for about 0.8 s raise a microsleep regardless of minute-scale PERCLOS (Boyle et al., 2008). Observer keys, rest-stop stimulant codes, and near-miss marks are written without names and joined later by Unix time. Sensitivity is defined only on paired rows labelled checking.

Result:

No licensed tanker night was labelled for this manuscript. On 120 synthetic 45-second sessions the fusion layer recovered the generating class. A separate 6.5-hour synthetic Tema–Kumasi template (seed 42) was used only to exercise rostering tables: 128 banners, gone rate 14.1 percent, PERCLOS 95th percentile crossing 0.30 after hour 4 even with ataya, depot cut 0.45 at TPR 0.86 and FPR 0.12, trunk cut 0.55 at TPR 0.68 and FPR 0.02, and startle-window mesh loss of 94–97 percent. Those figures are labelled SYNTHETIC. They are not

field performance.

Conclusion:

The contribution that does not require a fabricated accuracy is the specification: a detector that can be audited on a Monday CSV, that refuses to guess when the face disappears, and that treats PERCLOS percentiles as rostering input rather than as a charge sheet (Gander et al., 2011; Dawson et al., 2011). A coded night on one corridor remains the next measurement.

Keywords: driver fatigue; Eye Aspect Ratio; PERCLOS; facial landmarks; HAZMAT; West Africa

Dedication

This work is dedicated to my family, and to the commercial and tanker drivers whose working hours sit behind the occupational papers cited in Chapter Three.

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List of Abbreviations

Abbreviation	Meaning
ADR	Agreement concerning the International Carriage of Dangerous Goods by Road
EAR	Eye Aspect Ratio
EMA	Exponential moving average
FPR	False positive rate
HAZMAT	Hazardous materials
HGV	Heavy goods vehicle
KSS	Karolinska Sleepiness Scale
MAR	Mouth Aspect Ratio
PERCLOS	Percentage of eyelid closure over time
PLCDB	Proportion of long-closure-duration blinks
P80	PERCLOS with an 80 percent closure criterion
PVT	Psychomotor vigilance task
TPR	True positive rate
WA-PERCLOS-HYS	West Africa PERCLOS detector with hysteresis

Chapter One Introduction

1.1 Background and Motivation

Most open drowsiness repositories begin with a face, compute an Eye Aspect Ratio, and sound a file when the ratio stays low for a dozen frames. That exercise is legitimate in a classroom. It is a poor description of a loaded tanker at 02:14 on the Nsawam stretch. The cab is dark. Oncoming lights bloom a cheap sensor. Sunglasses from the afternoon heat sit on the dash. The phone that hosts the model is also the dispatcher line. If the model shouts at a talking mouth it will be taped over. If it stays quiet through a two-second closure the result is not a bruised bumper.

The World Health Organization (2023a) put annual road traffic deaths at 1.19 million in 2021, still the leading killer of people aged 5–29, and still concentrated in low- and middle-income countries. The Africa Status Report on Road Safety estimated 259,601 deaths on the continent in 2021, about a quarter of the world total (SSATP, 2025). The WHO African Region report for the same period described a fatality rate near 19.4 per 100,000 and a 17 percent rise in deaths since 2010, against a global trend that was at least no longer rising (World Health Organization, 2023b). Adeloye et al. (2016) had already described the African burden as under-counted in official returns. Inside Ghana, speed and injury patterns on trunk roads have been documented for two decades (Afukaar, 2003; Damsere-Derry et al., 2010; Mock et al., 2004).

Occupational surveys of heavy-goods and tanker drivers in Tema treat fatigue driving as an outcome of job demands, insecurity, and burnout (Amoadu & Akoto-Buabeng, 2025). Long-distance bus work in the same country shows the same pattern: long duty, irregular shifts, and sleepiness sitting on the path to crashes and near-misses (Amoadu et al., 2024). Dotse et al. (2019) sat with commercial passenger drivers and heard 18-to-20-hour days. Nævestad et al. (2025) compared Ghanaian and Norwegian bus drivers and found Ghanaian drivers far more likely to have dozed at the wheel; country remained a dominant term after work stress and safety culture were in the model. That is not a genetic claim. It is a claim about the road-labour regime.

Naturalistic work from richer fleets is not a substitute for those surveys, but it fixes an effect size when drowsiness can be observed. The 100-Car study put drowsy driving at a four- to six-fold increase in near-crash and crash risk (Klauer et al., 2006; Dingus et al., 2006). Connor et al. (2002) put the odds of an injury crash at 8.2 when the driver reported being sleepy, and attributed about 19 percent of injury crashes to acute sleepiness factors. Horne and Reyner (1995) had already shown that sleep-related accidents cluster at 02:00, 06:00 and 16:00, which is exactly the clock a night tanker runs. Pack et al. (1995) described the same

night-and-dawn shape in the United States. Folkard (1997) called those hours black times for transport safety.

Fatigue, in the sense used here, is the family of impairments that includes sleep restriction, time-on-task decline, circadian troughs, and the brief involuntary closures called microsleeps (Dinges, 1995; Phillips, 2015; Williamson et al., 2011). It is not interchangeable with boredom, and it is not interchangeable with alcohol, even though Dawson and Reid (1997) showed that sustained wakefulness can degrade performance on a scale that invites that comparison. Van Dongen et al. (2003) showed that chronic restriction accumulates even when people think they have adapted. Otmani et al. (2005) and Philip et al. (2005) showed that driving duration and partial sleep deprivation degrade subsequent alertness. Thiffault and Bergeron (2003) showed that a monotonous road accelerates the decline. A petroleum corridor after midnight is that road.

1.2 Statement of the Problem

West African bulk-road cabins do not match the assumptions of most published driver-monitoring stacks. Palpebral fissure height varies. Cabin luma collapses after dusk. Shared handsets break a calibration that was collected on another face. Stimulants such as ataya change the appearance of alertness without restoring sleep. A near miss can throw the face out of a mesh that was stable a second earlier. A single global EAR tripwire of 0.21 does not survive those conditions (Soukupová & Čech, 2016; Ersoy et al., 2026).

The problem of this thesis is therefore not to invent a new law of eyelid geometry. It is to specify a personally calibrated, hysteresis-based, multi-cue detector that can live in that cabin, that can be audited as numbers rather than as a softmax, and that does not present foreign dataset accuracies as Ghanaian field performance. Albadawi et al. (2022) warned, correctly, that dataset accuracy is not field accuracy. This study takes that warning as a design constraint.

1.3 Aim and Objectives

The aim is to design, implement, and bound a personally calibrated eye-state and facial-landmark fatigue detector for HAZMAT drivers in West Africa, with an evaluation that does not present foreign or synthetic numbers as local on-road accuracy.

The specific objectives are as follows.

First, to review ocular, vehicle, and physiological measurement families and retain only those cues that still run on a dashboard handset (Sahayadhas et al., 2012; Kolus, 2024).

Second, to specify a two-threshold hysteresis rule and a personal eight-second calibra-

tion so that landmark jitter does not chatter the alert.

Third, to implement the specification as inspectable software with nameless field logs and Unix-time pairing.

Fourth, to verify the implementation on synthetic sessions and to write the field protocol that a later coded night must keep.

1.4 Research Questions

Which ocular cues have an independent evidential base, and which of those cues still run on a dashboard handset?

How should closure be decided so that landmark jitter does not chatter the alert?

What personalisation is required when palpebral fissure height and cabin luma both vary?

What can a study without a labelled local video corpus actually claim?

1.5 Scope and Delimitations

The study is vision only. One forward RGB camera is assumed. There is no EEG channel, no steering-torque channel, and no lane tracker. Evaluation uses synthetic trajectories and a synthetic 6.5-hour haul template. No claim is made that those sessions equal a licensed night tanker. No claim is made that MediaPipe is fair across every West African face until a local corpus exists. In-cab video, where collected later, is to be discarded after coding. Logs carry landmarks, scores, Unix times, stimulant codes, and checking or gone labels. They do not carry names.

1.6 Significance of the Study

A tanker that drifts while the driver is mid-blink is a fire, a blocked trunk road, and a depot that cannot move product. The European Agreement concerning the International Carriage of Dangerous Goods by Road exists because a tank of petrol is not a crate of tomatoes (United Nations, 2023). Fatigue risk management belongs at the organisational level (Gander et al., 2011; Dawson et al., 2011). A camera that cannot be audited will not be adopted. A camera used as a disciplinary instrument is unlikely to be accepted. The significance of this study is that it writes the detector as a rostering input and as a public specification, not as a black box sold against an unpublished Ghanaian accuracy.

1.7 Organisation of the Thesis

The thesis is organised into nine chapters. Chapter One introduces the study. Chapter Two reviews fatigue measurement and driver monitoring. Chapter Three defines the West African HAZMAT context. Chapter Four presents the research methodology, and Chapter Five specifies the WA-PERCLOS-HYS algorithm. Chapter Six describes the implementation. Chapter Seven reports the synthetic verification results, Chapter Eight discusses their meaning and limitations, and Chapter Nine concludes with recommendations for field evaluation.

Chapter Two Literature Review

2.1 Sleepiness as a Crash Mechanism

Dinges (1995) reviewed sleepiness and accidents when the field still had to argue that the problem was real. Lyznicki et al. (1998) put the same argument in a medical journal. Maycock (1997) linked sleepiness to driving in a British sample. Sagberg (1999) added self-report of falling asleep at the wheel. Philip and Åkerstedt (2006) treated transport sleepiness as a public-health object rather than a curiosity. Williamson et al. (2011) tried to pin down how fatigue causes crashes, which is a harder question than whether it does.

The professional-driver subset is not safer by default. Häkkinen and Summala (2000) measured sleepiness in truckers. McCartt et al. (2000) did the same in the United States. Mitler et al. (1997) filmed truck drivers falling asleep on scheduled runs in a study that should have ended the argument that professional drivers are immune. Howard et al. (2004) linked sleepiness, sleep-disordered breathing, and accident risk in commercial vehicle drivers. Folkard and Tucker (2003) reviewed shift work, safety, and productivity. Anund et al. (2008) related night impairment to physiological sleepiness. Ingre et al. (2006) tied Karolinska scores to lane excursions. Jackson et al. (2013) separated cognitive components of simulated driving under sleep loss.

Connor et al. (2002) moved from counts to attributable risk: five hours of sleep or less, driving between 02:00 and 05:00, and a high Stanford score each raised injury-crash odds after confounders. Klauer et al. (2006), using instrumented vehicles rather than police forms, put drowsiness in the same causal neighbourhood as distraction. Those papers were not written in Accra. They still constrain what a West African detector is allowed to ignore. A night haul is not an edge case. It is the exposure.

2.2 Subjective Scales

The Karolinska Sleepiness Scale (Åkerstedt & Gillberg, 1990) and the Epworth scale (Johns, 1991) are useful in trials. A driver will not fill them in at 90 km/h. They belong in the field protocol as observer and rest-stop instruments, not in the cabin loop. Self-ratings also lag impairment (Dinges & Grace, 1998; Van Dongen et al., 2003). That lag is one reason PER-CLOS was preferred to introspection when the two were compared against the psychomotor vigilance task.

2.3 Physiological Sensors

Lal and Craig (2001) reviewed EEG markers of driver fatigue. Borghini et al. (2014) put EEG, heart, and eye measures into a single human-factors frame. Vicente et al. (2016) used heart-rate variability. Minhas et al. (2025) still needed EEG to audit a visual PERCLOS rule in drivers with obstructive sleep apnoea. The accuracy of those markers is not the issue in Tema. The wearability is the issue. Sahayadhas et al. (2012) said the quiet part: hybrids that include physiology work in laboratories and fail the adoption test.

2.4 Vehicle-Based Measures

Steering entropy and lane standard deviation move after control has already slipped (Sahayadhas et al., 2012; Sikander & Anwar, 2019). Wierwille et al. (1994) spent three years trying to make vehicle-based algorithms talk to rater scores and still found the face more informative than the wheel. For a 45,000-litre load the late vehicular signal is the wrong moment. This thesis therefore treats vehicular signals as a future complement, not as the primary channel. The Federal Motor Carrier Safety Administration (2011) hours-of-service rule is the organisational cousin of that late signal: it limits exposure; it does not watch the eyelid.

The Federal Motor Carrier Safety Administration (2011) hours-of-service rule limits exposure. It does not watch the eyelid. Dawson et al. (2011) and Gander et al. (2011) describe fatigue risk management as an organisational system. Belenky et al. (2003) and Van Dongen et al. (2003) describe the dose-response of sleep restriction. Balkin et al. (2011) describe the temptation and the limit of technological approaches. This detector sits inside that literature as one sensor, not as a replacement for duty limits. A fleet that uses WA-PERCLOS-HYS to justify an eighteenth hour has inverted the thesis.

Reyner and Horne (2000) showed that caffeine can blunt early-morning sleepiness for a while. Ataya in this protocol is logged for the same reason: a stimulant changes the appearance of the face and the risk of the next hour. It does not rewrite the last six hours of restriction. The p95 table is split by stimulant so that a roster clerk cannot pretend the tea was a second night of sleep.

2.5 PERCLOS and Blink Timing

PERCLOS earned its centre the hard way. Wierwille and Ellsworth (1994) showed that trained raters could score drowsiness from faces. Wierwille et al. (1994) then spent three years on vehicle-based algorithms. Dinges and Grace (1998) put PERCLOS against the psychomotor vigilance task and against the other candidate gadgets of the mid-1990s. PERCLOS tracked PVT lapses better than the competitors and better than the subjects' own sleepiness

ratings. The P80 definition: time the eyelid covers at least 80 percent of the pupil: is the version they defended.

Bergasa et al. (2006) put PERCLOS next to blink frequency, closure duration, nods, face position, and gaze, then fused them with a fuzzy rule. That list is still the right list. Quiles-Cucarella et al. (2024) recomputed EAR, MAR, PERCLOS, gaze, and head pose on a Raspberry Pi. Wu et al. (2024) tried to detect face and landmarks in one cheap network. Kolus (2024) found that eye-activity papers still lean on PERCLOS even when the rest of the stack is a neural net.

Long closures are not ordinary blinks. Caffier et al. (2003) and Schleicher et al. (2008) treated blink duration as a drowsiness correlate. Ordinary blinks sit near 0.10–0.30 s in alert driving. Boyle et al. (2008) described microsleeps as a separate, shorter, more dangerous object than a slow PERCLOS climb. The hard 0.8-second override in this thesis exists because of that distinction.

2.6 Eye Aspect Ratio and Landmarks

Soukupová and Čech (2016) showed that six points are enough to score openness if the points are good. Kazemi and Sullivan (2014) made the 68-point predictor fast enough for a laptop. King (2009) packaged the tooling. Viola and Jones (2001) had made the face box cheap a decade earlier. The genealogy is not mysterious. What failed in the wild was the habit of shipping one EAR threshold for every face. Personalised EAR and MAR thresholds, calibrated before the trip, are now an explicit design in recent work (Ersoy et al., 2026). This thesis takes that idea and adds hysteresis so that the calibrated number is not used as a hair-trigger.

Mouth geometry is weaker. Abtahi et al. (2014) released YawDD because talking and yawning share a camera. Danisman et al. (2010) and Alioua et al. (2012) tried to separate them. MAR in this system is a supporting channel with a consecutive-frame gate, not a siren of its own. Flores et al. (2010) warned in real time from visual information and still needed more than a single mouth cue.

Dinges and Grace (1998) did not crown EAR. They crowned PERCLOS against the psychomotor vigilance task. EAR is a cheap per-frame stand-in for the eyelid coverage that raters and PERCLOS cameras were scoring in the 1990s (Wierwille & Ellsworth, 1994; Wierwille et al., 1994). Treating a single EAR sample as a fatigue diagnosis inverts that hierarchy. The hysteresis gate exists so that EAR can become a duration. PERCLOS exists so that the duration can become a minute. The 0.8-second override exists so that a minute-scale average cannot hide a microsleep (Boyle et al., 2008).

Kolus (2024) found that eye-activity papers still lean on PERCLOS even when the rest of the stack is a neural net. That finding is the reason this implementation keeps a CSV a safety officer can read. A softmax that beats NTHU-DDD by two points does not tell a Tema

roster clerk which haul hour to split.

2.7 Dense Meshes on Phones

Kartynnik et al. (2019) described a 468-point mesh that runs on mobile GPUs. Grishchenko et al. (2020) put attention on eyes and lips. Bazarevsky et al. (2019) supplied the BlazeFace box that feeds the mesh. Deng et al. (2020) is the heavier alternative if a project can afford RetinaFace. This project cannot, not if the device is the driver's own phone. Jabbar et al. (2020) showed that a tiny CNN on landmarks can live on Android. Reddy et al. (2017), Park et al. (2016), and Dwivedi et al. (2014) showed the same for raw-pixel nets on public sets. Those nets are the documented fallback if landmarks die. They are not the primary path because a CSV of EAR and PERCLOS can be read on a Monday morning and a softmax cannot.

2.8 Datasets That Are Not West African Tankers

NTHU-DDD (Weng et al., 2016) is the set everyone quotes. It is simulated, mixed-ethnicity, glasses and night included, and still not a petroleum tanker on the Accra–Kumasi road. YawDD is a parked car (Abtahi et al., 2014). DROZY is a laboratory multimodality set (Massoz et al., 2016). Using those accuracies as Ghanaian field numbers would be a category error. This thesis cites them as engineering context and stops there.

2.9 Multi-Cue Fusion

Ji et al. (2004) fused eyelid, gaze, head, and expression, with infrared helping at night. Bergasa et al. (2006) fused six visual parameters. Quiles-Cucarella et al. (2024) fused EAR, MAR, PERCLOS, gaze, and pose. Ramzan et al. (2019) surveyed the same instinct. Tadesse et al. (2014) used a hidden Markov model for temporal structure. The lesson is stable: one cue lies, several cues argue. Chapter Five writes the argument down as weights a fleet can change.

2.10 Summary of the Review

The evidential core is PERCLOS against PVT, blink duration against sleepiness, EAR as a per-frame proxy that must be personalised, and fusion as insurance against a single lying channel. The adoption core is a phone, not an EEG cap, and a number a safety officer can read. The missing core is a labelled West African tanker night. Chapter Three states why that

night looks different from the papers just cited. Chapter Five specifies the detector that can be taken into it without pretending the night has already been run.

Chapter Three West African HAZMAT Context

3.1 Freight Operations

West Africa moves fuel by road because pipelines and rail do not reach most inland markets. Articulated trucks carry the large majority of Nigerian surface freight for that reason. Ghana's bulk-road vehicles out of Tema and Takoradi are the downstream's moving inventory. A crashed tanker is therefore an energy event as well as a trauma event. The Tema–Kumasi corridor is used in this thesis as the default night template because it is long enough for time-on-task to matter, dark enough for luma to collapse, and busy enough that a blocked carriageway is a national logistics problem.

3.2 Occupational Measurements

Amoadu and Akoto-Buabeng (2025) surveyed 1,575 HGV drivers in Tema: 910 truck, 665 tanker. Psychosocial safety climate predicted less fatigue driving. Job insecurity, psychological demands, and burnout predicted more. Tanker drivers reported more job resources than truck drivers. The fatigue path did not disappear. A related Ghanaian bus sample recorded at least one crash in 53.7 percent of drivers over two years and a near-miss in 96.9 percent, with long hours and sleepiness among the risk terms (Amoadu et al., 2024).

Dotse et al. (2019) sat with commercial passenger drivers and heard 18-to-20-hour days, six or seven days a week, and no relief driver for sleepiness. Nævestad et al. (2025), comparing Ghanaian and Norwegian bus drivers, found Ghanaian drivers far more likely to have dozed at the wheel. Nigerian articulated-truck questionnaires on major freight routes have reported short rest in transit and trips longer than eleven hours, with travel duration associated with accident frequency. Commercial motor-tricycle drivers in Kumasi reported median ten-hour days and stimulant use associated with crashes. The mode is different. The hours are not.

3.3 Continental Road Safety Context

SSATP (2025) estimated 259,601 African road deaths in 2021. The WHO African Region report counted 225,482 deaths in the Region that year, 19 percent of the world total on about 3 percent of the world fleet, with a rate of 19.4 per 100,000 (World Health Organization, 2023b). The United Nations (2021) Global Plan for the Decade of Action for Road Safety 2021–2030 is the policy envelope. Ghana's National Road Safety Authority figures for recent years describe large administrative counts of crashes and deaths. Those are not a fatigue

attribution. They are the backdrop against which a missed microsleep has to be judged.

3.4 Cab Photometry

Night running is scheduled. Mean luma in the face box falls. Headlights create a dynamic range a phone meter was not designed for. A peaked cap throws a bar of shadow across the palpebral fissure. Sunglasses are rational after a day loaded in Tema heat. Ji et al. (2004) and Bergasa et al. (2006) solved the night problem with active infrared. This thesis cannot assume that illuminator. It can assume an eight-second calibration and a night-mode scale on the EAR threshold, and it can refuse to guess when the face disappears. Harmattan film on a cheap lens is a separate optics state, not a fatigue state. Dark skin in a poorly metered RGB frame is a photon problem before it is a model problem.

3.5 HAZMAT Consequences

The ADR exists because dangerous goods multiply the cost of an ordinary crash (United Nations, 2023). ECOWAS and national dangerous-goods rules try to import that logic. They do not put a camera in the cab. The loss function of a false negative is therefore not symmetric with a passenger-car demo. The loss function of a false positive that startles a driver onto a soft shoulder is not zero either. Chapter Five's cooldown, two operating points, and spoken escalation exist because of that asymmetry.

Afukaar (2003), Damsere-Derry et al. (2010), Mock et al. (2004), and Adeloye et al. (2016) are not eyelid papers. They are cited because a drowsy tanker on those roads is not a fender-bender. The Decade of Action plan (United Nations, 2021) and the ADR (United Nations, 2023) are cited for the same reason. A computer-vision thesis that ignores consequence will pick the wrong operating point. Chapter Five's two cuts exist because those papers exist.

3.6 Design Constraints

The cabin writes the requirements. Offline first. Calibrate at the gate. Prefer numbers a safety officer can read. Emit NO_FACE rather than an unsupported fatigue classification. Log events without uploading a night of identifiable video. Treat the Monday CSV as rostering information, not as a charge sheet (Dawson et al., 2011; Gander et al., 2011).

Table 3.1 Design constraints for the West African cabin.

Constraint	Design answer
Dark cab, no IR assumed	Night luma scale 0.92; refuse when face lost
Shared phone	Face hash at the gate; SHARED_DEVICE state
Sunglasses after dusk	DEGRADED state; no fatigue guess
Harmattan film / dirty lens	OPTICS_DIRTY from eye-crop sharpness
Monocular / one dead eye	Both-eye rule with monocular fallback
Talking on radio	Yawn channel gated; talking flag
Ataya at rest stop	Stimulant code on rest_stops.csv; p95 split
Near miss throws the mesh	30 s startle clip; report blind fraction
Informal contracts	Nameless logs; ban on sole-source dismissal
Dead radio / no cloud	On-device inference; local CSV

Chapter Four Research Methodology

This chapter defines the research design, available data, frame-level measurements, temporal features, decision rules, evaluation units, ethics, and reproducibility protocol. Parameter values are taken from configs/default.yaml in the public repository. Any later change to that configuration should be reflected in this chapter before submission.

4.1 Research Design

The study uses a specification-and-verification design rather than supervised learning on a labelled Ghanaian corpus. There is no local tanker video. There is therefore no train–validate–test split of faces. The primary artefact is a deterministic pipeline whose every cut can be read in YAML. The secondary artefact is a pair of synthetic objects that ask whether that pipeline implements its own rules. The tertiary artefact is a field protocol that a later night must keep if Question B: does this cut tanker fires on Tema–Kumasi: is ever to be answered.

Sahayadhas et al. (2012) and Sikander and Anwar (2019) split fatigue sensing into vehicle, behavioural, and physiological families. This design chooses the behavioural family only, on one forward RGB camera, because an EEG cap and a torque sensor fail the adoption test stated in Chapter Three. Albadawi et al. (2022) warned that dataset accuracy is not field accuracy. The design takes that warning as a prohibition: NTHU-DDD and YawDD numbers are not reported as Tema results (Weng et al., 2016; Abtahi et al., 2014).

The unit of analysis for the wiring test is a 45-second session after an 80-frame calibration. The unit of analysis for rostering is a haul hour on a 6.5-hour template. The unit of analysis for the gold set is a paired banner with label=checking. The unit of analysis for startle is a 30-second clip after a near-miss mark. These four units are not interchangeable and must be reported separately to avoid overstating accuracy.

4.2 Philosophical and Practical Stance

The study is instrumental. It does not claim that EAR is fatigue. It claims that eyelid geometry, aggregated the way Dinges and Grace (1998) aggregated it, is the precursor a phone can measure without electrodes (Lal & Craig, 2001; Borghini et al., 2014). Phillips (2015) reviewed definitions of fatigue and warned against collapsing sleepiness, time-on-task, and boredom into one word. This method keeps microsleep as a separate override precisely so that collapse does not happen inside the code (Boyle et al., 2008).

Gander et al. (2011) and Dawson et al. (2011) treat fatigue risk management as an organisational system. The method therefore produces a Monday CSV rather than a private

softmax. A detector that a safety officer cannot explain is unlikely to be trusted or retained. That sentence is a methodological constraint, not a slogan.

4.3 Population Sample and Field Data Limitation

The intended population is HAZMAT and heavy-goods drivers on West African night corridors, with Tema–Kumasi as the default (Amoadu & Akoto-Buabeng, 2025; Amoadu et al., 2024; Dotse et al., 2019). The achieved sample for this manuscript is zero licensed drivers. That fact is stated in the abstract and again here so that it cannot be moved to a footnote.

What stands in for a sample is two synthetic objects. Object A is 120 sessions of 45 seconds at 25 frames per second: 40 alert, 40 drowsy, 40 microsleep, plus an 80-frame open-eye warm-up used only for the gate. Object B is a 6.5-hour Tema–Kumasi night template with seed 42, a rest-stop ataya mark at hour 2.05, three near-miss marks, lognormal observer lag, and a 1 Hz PERCLOS series. Object A tests `fusion.py` against the algorithm specification in Chapter Five. Object B tests `field_protocol.py`, `pairing.py`, `monday_sheet.py`, `analyze_haul.py`, and `overwrite_rostering.py` against Chapter Four section 4.10. Neither object is a driver.

4.4 Materials and Apparatus

The intended apparatus is a dashboard handset running `scripts/run_detector.py`, OpenCV, and MediaPipe Face Mesh with `refine_landmarks true`, `min_detection_confidence 0.6`, `min_tracking_confidence 0.5`, `max_num_faces 1` (Kartynnik et al., 2019; Grishchenko et al., 2020; Bazarevsky et al., 2019). Camera defaults in `configs/default.yaml` are `index 0`, `640 by 480`, `fps_hint 25`. Those defaults match a cheap phone preview, not a research rig.

The observer apparatus is the same window. Keys `c`, `g`, `s`, `1–9`, `n`, and `q` are handled only when that window has focus. This is an operational constraint of the OpenCV interface. The coding machine after the night keeps numeric CSVs and deletes raw video (Appendix C).

Infrared illuminators used by Ji et al. (2004) and Bergasa et al. (2006) are not in the bill of materials. They are listed in Chapter Seven as a recommendation for a funded trial. Putting them in the method section as if they were used would be false.

4.5 Live Night Procedure

Step one. At the depot gate the driver faces the boom for eight seconds. `adaptive.calibration_seconds` is 8. `adaptive.min_valid_frames` is 80. Face required. EMA

already running at `metrics.ema_alpha 0.4`. `EAR_open` is the median of samples at or above the 60th percentile of that window. The current implementation finalises after eight seconds when at least $\max(15, \min_valid_frames/4)$ valid samples are available. With the default `min_valid_frames` value of 80, this is 20 samples. It rejects `EAR_open` below `adaptive.ear_open_min 0.18` or above `adaptive.ear_open_max 0.45`.

Step two. Luma in the face box is scored as Rec. 601. If the median is under `adaptive.night_mode_luma_threshold 70`, the night scale `adaptive.low_light_ear_scale 0.92` is applied. θ_close is `clip(adaptive.close_ratio 0.68  $\times$  s  $\times$  EAR_open, 0.16, 0.26)`. θ_open is `min( $\theta\_close$  + metrics.ear_hysteresis_gap 0.03, EAR_open - 0.02)`. `identity.enable` hashes the gate face. A later mismatch beyond `identity.mismatch_l2 28` raises `SHARED_DEVICE`.

Step three. On the road, each frame yields `EAR`, `MAR`, `pitch`, `optics sharpness`, and a talking flag if `voice.energy_threshold 0.035` is crossed. `HysteresisGate.step` updates `closed`. `PERCLOS` is the mean of `closed` over `metrics.perclos_window_seconds 60`. A closed run of `metrics.consecutive_closed_frames 20` at 25 FPS is 0.8 s. `MAR` at or above `metrics.mar_yawn_threshold 0.60` for `metrics.yawn_consecutive_frames 12` is a yawn run unless talking is set. `Nod` uses `metrics.nod_pitch_degrees 20` unless `imu.rough_rms 2.4` says the road is bouncing.

Step four. After a `DROWSY` or `MICROSLEEP` banner the observer hits `c` or `g`. At a rest stop the observer hits `s` and a Karolinska key (Åkerstedt & Gillberg, 1990). After a near miss the observer hits `n` so that the later analysis can evaluate the following 30-second window. `Alert-label` and `rest-stop` rows store `alert_unix_time`. No names.

Step five. After the night, `pairing.py` joins labels to `alerts.csv`. `monday_sheet.py` prints counts. `overwrite_rostering.py` rebuilds the workbook. Raw video is deleted. The cover banner `SYNTHETIC` is replaced only when these steps used a real tanker.

At the Tema gate the driver sits, looks at the boom, and does not talk for eight seconds. The HUD shows `CALIBRATING`. If fewer than 20 valid samples arrive after eight seconds under the current default configuration, calibration remains incomplete and should be repeated. If `EAR_open` is under 0.18 the officer looks for sunglasses or a failed mesh. If the face hash later changes, `SHARED_DEVICE` is raised and the night is no longer a fatigue night until the gate is repeated.

On the road the observer keeps the OpenCV window focused. After every banner the observer hits `c` or `g` before the next event. At a rest stop the observer hits `s` until the stimulant code matches what was consumed and hits a Karolinska key. After a near miss the observer hits `n`. At the depot the officer runs `monday_sheet.py` and `overwrite_rostering.py`. Raw video is deleted after coding. The CSVs stay. These steps form the field protocol and should be followed consistently.

4.6 Preprocessing

Raw pixels are not stored as the product. MediaPipe returns 468 landmarks. Six points per eye become EAR after Soukupová and Čech (2016). Left indices are 33, 160, 158, 133, 153, 144. Right indices are 362, 385, 387, 263, 373, 380. The mean is smoothed with EMA $\alpha = 0.4$ before any threshold. Landmark jitter of a pixel on a twenty-pixel eye is enough to chatter a single cut (Quiles-Cucarella et al., 2024). Smoothing is therefore preprocessing, not fusion.

Mid-haul open-eye re-estimation is allowed after `adaptive.midhaul_seconds` 2700 but clamped to `adaptive.midhaul_clamp` 0.03 of the gate value. Slow sag appears as drift on the HUD rather than as a silent rewrite of θ_{close} . That clamp is the difference between calibration and contamination.

4.7 Feature Construction

Features were taken from the measurement literature, not from a private grid search on a hidden test set. EAR is the per-frame proxy (Soukupová & Čech, 2016; Ersoy et al., 2026). PERCLOS is the backbone (Wierwille et al., 1994; Dinges & Grace, 1998; Kolus, 2024). PLCDB uses `metrics.long_blink_seconds` 0.40 (Caffier et al., 2003; Schleicher et al., 2008). MAR uses 0.60 and a 12-frame gate because talking and yawning share a camera (Abtahi et al., 2014; Danisman et al., 2010; Alioua et al., 2012). Pitch is a two-dimensional nod hint, weight 0.10, discarded on a rough road.

Oversampling is not used. There is no imbalanced labelled classifier. Class balance on Object A was imposed by construction: forty sessions in each generating class. Applying SMOTE to a specification study would pretend that a sampling trick can stand in for a missing field corpus.

4.8 Decision Rules

HysteresisGate stores closed as a boolean. If closed is false and both raw eye EAR values are below θ_{close} , closed becomes true. If closed is true and the smoothed mean EAR is above θ_{open} , closed becomes false. Both eyes must agree unless the monocular fallback identifies one eye as unusable. Evidence maps and weights are exactly the YAML block `fusion.weights`: `perclos` 0.35, `ear_closure` 0.25, `long_blink` 0.15, `yawn` 0.15, `head_nod` 0.10.

Table 4.1 Method parameters from `configs/default.yaml`.

Parameter	Default
EAR close (pre-calibration)	0.21
Calibrated close threshold	$0.68 \times \text{open-eye median}$, band [0.16, 0.26]

Parameter	Default
Hysteresis gap	0.03
EMA alpha	0.40
Night luma floor	70
Night scale	0.92
PERCLOS window	60 s
PERCLOS map	0.30
Consecutive closed frames	20 (0.8 s at 25 FPS)
Long blink	0.40 s
MAR yawn	0.60
Depot / drowsy score	0.45
Trunk score	0.55
Severe score	0.70
Cooldown	4 s
Observer pairing window	120 s
Rest-stop pairing window	2 h
Startle clip	30 s
Mesh-lost floor	0.45
Mid-haul open-eye clamp	± 0.03
Calibration valid-sample behaviour	min_valid_frames = 80; current code finalises after 8 s with at least 20 valid samples

ALERT if the score is under the drowsy cut and the closed run is under 20. DROWSY if the score is at least the drowsy cut and under 0.70. MICROSLEEP if the score is at least 0.70 or the closed run reaches 20. `west_africa.operating_point` selects `depot_road` (0.45) or `trunk_road` (0.55). `alerts.escalate_after` 3 repeated banners escalates the spoken line. None of those rules is learned from Object A.

4.9 Refusal States

CALIBRATING, NO_FACE, OPTICS_DIRTY, DEGRADED, and SHARED_DEVICE do not emit a fatigue guess. DEGRADED is glasses after `optics.sunset_hour_local` 18. OPTICS_DIRTY is eye-crop sharpness under `optics.sharpness_min` 18. SHARED_DEVICE is a face-hash mismatch. Preferring a refusal state to an unsupported fatigue classification is part of the method. A development run with heavy dropouts showed why: drowsy and microsleep sessions collapsed toward ALERT when the mesh data were low quality. Ji et al. (2004) addressed this failure mode with infrared. This method records low-quality mesh data as a refusal state.

4.10 Pairing Gold Set and Rostering Metrics

pairing.py uses alert_unix_time as a hint and selects the nearest alert within 120 seconds. If the hint is absent, it selects the nearest prior alert within 120 seconds for labels. Rest stops use the nearest prior alert within two hours. Sensitivity uses only paired=yes and label=checking. The gone rate is n_gone / n_banners and is printed on the same sheet. Observer lag is keypress Unix time minus paired alert Unix time. PERCLOS p95 by haul hour, split by the stimulant on the last rest-stop row, is the rostering input, not an accuracy. Startle-loss is the fraction of the 30 s after a near-miss mark in which mesh confidence stays under 0.45.

No inferential test is reported as a field finding. Mixed-effects models with driver as a random intercept belong to the protocol that has not been run (Ingre et al., 2006). The synthetic ROC uses a latent fatigue process plus $N(0, 0.10)$ so that detector state is not both score and truth.

An observer who was not looking cannot corroborate a banner. Counting those banners as true positives inflates sensitivity. Counting them as false positives punishes the detector for an observer failure. The only honest treatment is to drop them from the gold set and to print the gone rate next to the sensitivity. Fourteen percent gone on the synthetic night is already large enough to move a claimed accuracy. A field night that hides the gone rate is not a field night this protocol will accept.

The pairing window of 120 seconds is a compromise between an observer who keys immediately and an observer who finishes a radio call. Median checking lag on the synthetic night was 8.3 s. A later real night should publish its own lag histogram with plot_deltas.py. That histogram is an operational metric, not a model metric.

4.11 Ethics

Consent language states purpose as fatigue-alert research, not criminal investigation. One forward camera. Keep landmarks and scores. Delete raw video after coding. No names, plates, or phone numbers. Face hash only at the gate. Ninety-day retention of numeric logs unless a lawful investigation requires longer. Right to refuse the camera and still take the load. Refusal is not stored as a name. Ban on using the log as the sole ground for dismissal (Gander et al., 2011; Amoadu & Akoto-Buabeng, 2025). Those rules are in Appendix C because they are part of how data are allowed to exist, not because they are decoration after the results.

In-cab video of a hired driver is identifiable. Landmark files are less so and still need a retention limit. Ninety days of numeric logs is the default. Named officers and named researchers are the only readers. Refusal of the camera is allowed and is not stored as a name. A banner is not a disciplinary event. Using the Monday sheet as the sole ground for dismissal is banned because the detector has not been shown a Tema night and because burnout is an

organisational fact in the Tema surveys (Amoadu & Akoto-Buabeng, 2025). A system used primarily for discipline is unlikely to retain driver acceptance, whereas one used to inform rostering has a better chance of continued use.

Consent language states purpose as fatigue-alert research, not criminal investigation. That sentence is copied onto the ethics sheet in Appendix C so that a depot manager does not have to invent it at 18:00 when the truck is already late.

4.12 Reproducibility

`pytest tests -q` must pass on `test_hysteresis.py`, `test_metrics.py`, and `test_pairing.py` before anyone treats a cab log as a result. `scripts/evaluate.py` runs Object A. `scripts/overwrite_rostering.py` rebuilds the workbook from a logs folder. Seed 42 rebuilds Object B. Independent replication starts there, not with an unpublished video and an unpublished accuracy.

Clone the repository. Run `pytest`. Run `evaluate.py`. Confirm that HysteresisGate holds through a noisy rising edge. Confirm that pairing joins on `alert_unix_time`. Generate or load a logs folder. Run `overwrite_rostering.py`. Confirm that the cover sheet still says SYNTHETIC until a real night replaces it. Only after these checks should the threshold choice $\theta_{\text{close}} = 0.68$ be evaluated. A replication that begins with an unpublished accuracy on an unpublished video is not a replication of this thesis.

Chapter Five WA PERCLOS HYS Algorithm

5.1 Algorithm Foundations

The algorithm integrates established components into a single auditable pipeline. The face box is BlazeFace (Bazarevsky et al., 2019). The mesh is Face Mesh and Attention Mesh (Karttynnik et al., 2019; Grishchenko et al., 2020). Openness is EAR (Soukupová & Čech, 2016). Aggregate closure is PERCLOS in the P80 tradition (Wierwille et al., 1994; Dinges & Grace, 1998). Long blinks follow duration statistics (Caffier et al., 2003; Schleicher et al., 2008). Yawns use MAR with a run-length gate (Abtahi et al., 2014). Fusion is multi-cue (Ji et al., 2004; Bergasa et al., 2006). Personal thresholds are pre-trip calibration (Ersoy et al., 2026). What this study adds is the combination required by the cab: exponential smoothing, two-threshold hysteresis, both-eye agreement, night-luma scaling, a microsleep override separate from PERCLOS, two consequence-weighted cuts, and an alert policy that can be audited.

5.2 Per-Frame Geometry

Let the six eye points follow Soukupová and Čech (2016). EAR is the sum of the two vertical distances divided by twice the horizontal corner distance. Left and right EAR are computed on MediaPipe indices (33, 160, 158, 133, 153, 144) and (362, 385, 387, 263, 373, 380). The mean EAR is smoothed by mixing the current value with the previous smoothed value at $\alpha = 0.4$. In the current pipeline, both raw eye EAR values must cross the close threshold, while the smoothed mean controls release above the open threshold. Smoothing is not decoration. Landmark jitter of a pixel on a twenty-pixel eye width is enough to cross a single threshold twice in two frames. Quiles-Cucarella et al. (2024) and every demo that uses a consecutive-frame counter are fighting the same noise. EMA fights it before the counter starts.

MAR uses mouth corners and inner lips. Pitch is a two-dimensional surrogate from the eye-line-to-nose over eye-line-to-chin ratio. It is a nod hint, not a PnP solve. Rough-road bounce is a reason to discount the nod channel rather than to invent a three-dimensional head pose the phone cannot afford.

5.3 Hysteresis

A Schmitt trigger provides the two-threshold decision rule. The eye is closed when the smoothed EAR falls through θ_{close} and stays closed until it rises through θ_{open} , which is strictly larger than θ_{close} . The default gap is 0.03, clipped so that θ_{open} remains below the open-eye baseline by at least 0.02. Soukupová and Čech (2016) already showed that a

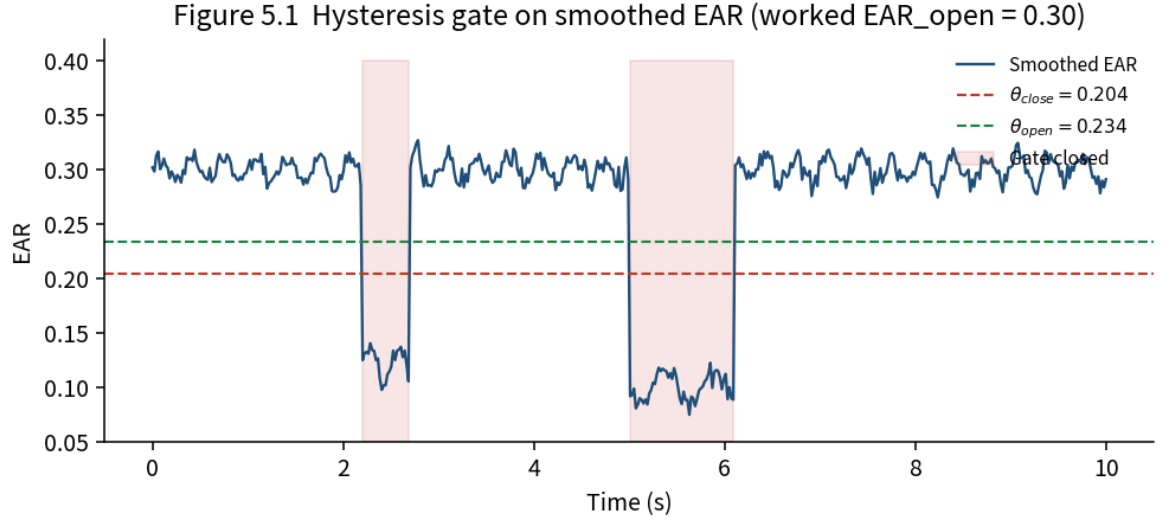


Figure 5.1 Schmitt trigger on smoothed EAR. Close at 0.204, open at 0.234 for EAR_{open} = 0.30.

single cut at 0.20 fires on half-blinks and on landmark wobble; they moved the decision into a short-window SVM for that reason. This thesis retains an interpretable rule-based decision suitable for on-device deployment. The two-threshold rule is the part of their lesson that still runs on a phone and still produces a duration that PERCLOS can add up.

When θ_{close} sits well below the open-eye cloud (the 0.68 rule on a mid-sized eye), Gaussian landmark noise almost never recrosses the cut. The gap’s job in that regime is to hold the closed state through the noisy rising edge of a real blink so that blink duration, not blink count, stays honest. When the cut is too close to the open-eye cloud (ratio ≥ 0.80 , or a fixed 0.21 on a small eye), the single threshold chatters. On those traces a 0.03 gap cuts open-eye flips. Gaps of 0.05 and above start to inflate alert-driving PERCLOS because the state lingers after the lid has already opened. The 0.03 default maintains the closed state through a noisy rising edge without excessively extending the closure. Both raw eye EAR values must cross the close threshold, reducing false closures from a wink, a pillar shadow, or one obscured lens.

5.4 Threshold Calibration

The dangerous number is not the gap. It is θ_{close} as a fraction of this driver’s open-eye EAR. A published constant of 0.21 is not a calibration. On a small-eye driver it sits inside the open-eye cloud: alert-frame F1 collapses and the state flips tens of times in tens of seconds. On a large-eye driver it sits too low: drowsy-closure recall falls. The constant performs acceptably near EAR_{open} ≈ 0.30 but can fail for smaller or larger open-eye baselines.

Half of baseline is the wrong objective for fatigue. A blink detector may want a 50 percent cut of a short open-eye window. Used as a fatigue close-threshold that cut under-calls

long closures. Blink detection and fatigue detection share EAR and do not share a threshold. The useful band for a fatigue close-threshold on the synthetic sweep used to tune this study was 0.65–0.72 of the high-band open-eye median. Combined score peaked at 0.68–0.72. The thesis keeps 0.68 because it is inside that peak and because it still leaves a visible gap between θ_{open} and EAR_{open} .

Do not estimate θ_{close} as $\mu - k\sigma$ from an eight-second open-eye window. That window is almost all open-eye samples. The resulting cut sits near the lower tail of the open cloud and chatters. Mean-minus-spread methods belong to protocols that have labelled closed frames. The gate does not. Eight seconds is enough if the baseline is a high-band median. Planting blinks in the gate window moved the P60-median close-threshold by less than 0.001 in contaminated calibrations. The P60 rule is insurance against a look-down or a yawn at the boom gate, not a claim that blinks wreck an eight-second median.

The procedure is fixed. Eight seconds at the depot gate, face required, EMA already running. EAR_{open} is the median of samples at or above the 60th percentile. The configuration stores $\text{min_valid_frames} = 80$. In the current implementation, calibration finalises after eight seconds when at least $\max(15, \text{min_valid_frames}/4)$ valid samples are available, which is 20 under the default configuration. Reject the calibration if EAR_{open} is under 0.18 (sunglasses or a failed mesh), or if EAR_{open} exceeds 0.45 (implausible landmarks). Luma scale s equals 0.92 if median Rec. 601 luma in the face box is below 70, else 1.0. Then $\theta_{\text{close}} = \text{clip}(0.68 \times s \times \text{EAR}_{\text{open}}, 0.16, 0.26)$ and $\theta_{\text{open}} = \min(\theta_{\text{close}} + 0.03, \text{EAR}_{\text{open}} - 0.02)$. Mid-haul re-estimates of the open-eye baseline are clamped to ± 0.03 of the gate value so that slow sag is visible as drift rather than silently rewriting the cut. Calibration on an already somnolent driver remains a residual risk. The protocol therefore puts the gate at the depot, not at kilometre 180.

Table 5.1 Worked hysteresis thresholds from the open-eye median.

EAR_{open}	θ_{close} (0.68)	θ_{open}	What a fixed 0.21 does
0.22	0.16 (clip)	0.19	Treats open eyes as closed
0.26	0.18	0.21	Borderline chatter
0.30	0.20	0.23	Looks fine, hides the problem
0.34	0.23	0.26	Misses some drowsy closures
0.38	0.26 (clip)	0.29	Misses more drowsy closures

A driver whose eight-second P60-median open-eye EAR is 0.30, with face luma 95, receives $\theta_{\text{close}} = 0.204$ and $\theta_{\text{open}} = 0.234$. Twenty frames of smoothed EAR at 0.19 hold MICROSLEEP after 0.8 s even if minute-scale PERCLOS is still 0.12. That is the Boyle

distinction written as a counter.

A driver whose open-eye EAR is 0.22, with face luma 48, receives the night scale 0.92 and then the clip at 0.16. A published constant of 0.21 would have treated that driver's ordinary blinks as closures. The clip is the admission that the linear 0.68 rule cannot be trusted below a small-eye floor.

A driver whose open-eye EAR is 0.38 receives θ_{close} clipped at 0.26. The large palpebral fissure would otherwise place the close cut so high that drowsy half-closures never trip. Personalisation without a clip still fails at the tails. Personalisation with a clip still needs a later infrared path when luma is the real problem.

These three windows are teaching arithmetic. They are not measurements from Tema. They exist so that a second implementer can check fusion.py against Chapter Five without guessing what 0.68 of 0.30 was supposed to be.

5.5 Temporal Features

Over a 60-second window at F frames per second, PERCLOS is the fraction of frames whose hysteresis state is closed. PLCDB is the fraction of completed blinks whose duration is at least 0.40 s. A consecutive-closure counter increments while the hysteresis state is closed and resets when it opens. Twenty frames at 25 FPS is 0.8 s and is treated as a microsleep candidate (Boyle et al., 2008). A yawn counter increments while MAR exceeds 0.60 and resets otherwise, or immediately if a talking flag is set. Twelve frames is the yawn gate.

5.6 Fusion and Operating Points

Evidence is mapped into $[0, 1]$ and weighted. PERCLOS is divided by 0.30 and carries weight 0.35. The closure run is divided by 20 frames and carries 0.25. PLCDB is divided by 0.35 and carries 0.15. The yawn run is divided by 12 and carries 0.15. Nod is $\max(0, \text{pitch degrees}) / 20$ and carries 0.10, unless a rough-road flag zeroes it.

Table 5.2 Evidence channels maps and fusion weights.

Channel	Map	Weight
PERCLOS	PERCLOS / 0.30	0.35
Closure run	$C_t / 20$	0.25
PLCDB	PLCDB / 0.35	0.15
Yawn run	yawn frames / 12	0.15
Nod	$\max(0, \text{pitch}^\circ) / 20$	0.10

ALERT if the score is under the drowsy cut and the closure run is under 20 frames.

DROWSY if the score is at least the drowsy cut and under 0.70. MICROSLEEP if the score is at least 0.70 or the closure run reaches 20 frames. The depot-road drowsy cut is 0.45. The trunk-road drowsy cut may be raised to 0.55 on the same night. That is a fleet choice written into configuration, not a second algorithm. Consequence is not symmetric: a false negative on the N6 with 45,000 litres is not the same event as a false positive leaving the Tema gate.

Leaving the Tema gate at walking speed is not the N6 at 70 km/h with a full tank. A depot cut of 0.45 accepts more false positives in exchange for earlier rest prompts. A trunk cut of 0.55 accepts more missed drowsy frames in exchange for fewer startle shouts. Both cuts are configuration. Neither cut is a second algorithm. The synthetic ROC in Chapter Seven is a latent process plus noise so that detector state is not used as both score and truth. A circular ROC would have been easier to draw and would have been a lie.

5.7 Alert Policy

Alerts have a four-second cooldown so that a flickering score does not become a flickering shout. Messages name the action: rest at the next safe lay-by, not a generic beep. Every issued alert writes Unix time, state, score, and channel names to alerts.csv. The spoken sentence is part of the method. A safety officer who cannot explain an alert is unlikely to trust or retain the system.

5.8 Optics Identity and Degraded States

OPTICS_DIRTY is raised when eye-crop sharpness falls below a floor. DEGRADED is raised when glasses are on after local sunset. SHARED_DEVICE is raised when the face hash does not match the gate face. NO_FACE is raised when the mesh is absent. CALIBRATING is raised during the eight-second gate. None of those states emits a fatigue guess. Monocular fallback is allowed when one eye is dead and the other is live. The flag is written so that a Monday sheet can see that the night was not binocular.

5.9 Field Logging

The observer in the jump seat is part of the algorithm's measurement theory. After a banner the keys are c for checking and g for gone. Rest-stop key s cycles the stimulant code. Keys 1–9 write a Karolinska score. Key n marks a near miss for later analysis of the following 30-second window. A 1 Hz row of PERCLOS, EAR, state, and mesh-present is written automatically. No names appear in any CSV. Each protocol row stores alert_unix_time from the last DROWSY or MICROSLEEP banner so that pairing does not depend on a shared clock

note.

5.10 Gold-Set Rule

Sensitivity is reported only on rows with paired=yes and label=checking. The gone rate is quoted on the same table. Untagged banners are neither gold nor hidden. That rule is the difference between a detector paper and a detector that can be argued with a safety officer who was not in the jump seat.

Chapter Six System Implementation

6.1 Package Layout

The package `fatigue_wa` splits geometry from the camera on purpose. `landmarks.py` is the only file that knows mesh indices. `metrics.py` holds EAR, MAR, PERCLOS, long-blink bookkeeping, and the EMA smoother. `adaptive.py` holds calibration, luma scale, and the mid-haul clamp. `fusion.py` holds `HysteresisGate` and `EvidenceFusion`. `alerts.py` holds cooldown and the spoken line. `optics.py` holds sharpness. `identity.py` holds the gate face hash. `field_protocol.py` writes observer keys, rest-stop stimulant and Karolinska scores, near-miss marks, and the 1 Hz series. `pairing.py` joins those rows to `alerts.csv` by Unix time. `pipeline.py` is the seam. Configuration lives in `configs/default.yaml` so that a fleet can change a cut without editing Python.

The public repository is named in Appendix A. Reproduction is `pytest`, `evaluate.py`, `run_detector.py`, `monday_sheet.py`, `plot_deltas.py`, `analyze_haul.py`, and `overwrite_rostering.py`. That list is the implementation claim. Each implementation claim in this chapter corresponds to a file in the repository.

`landmarks.py` converts MediaPipe landmarks into pixel coordinates and the six-point eye sets. It does not decide closure. `metrics.py` computes EAR, MAR, pitch, EMA, blink durations, and PERCLOS from the hysteresis boolean it is given. `adaptive.py` owns the eight-second gate, the P60 median, the luma scale, the clip, and the mid-haul clamp. `fusion.py` owns `HysteresisGate.step` and the weighted sum. `alerts.py` owns cooldown and the spoken line. `optics.py` owns Laplacian sharpness on the eye crop. `identity.py` owns a coarse face hash at the gate. `field_protocol.py` owns keys and the 1 Hz file. `pairing.py` owns Unix time. `pipeline.py` is the only file that calls all of them in order. A change to an index belongs in `landmarks.py`. A change to a cut belongs in `default.yaml`. That separation is the implementation contribution.

`tests/test_hysteresis.py` constructs a falling-then-rising EAR trace with Gaussian jitter and asserts that the closed flag does not chatter across the gap. `tests/test_metrics.py` asserts that a geometrically open six-point eye has EAR above 0.25 and that EMA lags a step change. `tests/test_pairing.py` writes a tiny `alerts.csv` and a label with an exact `alert_unix_time`, then a second label without that column, and asserts both joins. Those three files are the minimum a second student needs before touching the cab.

6.2 Live Detector and HUD

`scripts/run_detector.py` opens the camera or a video file, runs Face Mesh, and draws a HUD. The HUD prints STATE, EAR, MAR, PERCLOS, closed= (the Schmitt hold), and drift=

(open-eye sag relative to the gate). The key legend is on the same rectangle: c check, g gone, s stim, 1–9 KSS, n miss, q quit. The OpenCV window must remain focused for the keys to register.

6.3 Pairing and Monday Sheet

pairing.py uses alert_unix_time as a hint and selects the nearest alert within 120 seconds. If the hint is missing, it falls back to the nearest prior alert within 120 seconds for labels. Rest stops use the nearest prior alert within two hours. The join writes paired_alerts.csv and paired_rest_stops.csv with a delta_s column. monday_sheet.py prints counts a safety officer can read: banners, checking, gone, gone rate, median lag. No names are required for the join. No names are permitted in the files.

6.4 Rostering Workbook

overwrite_rostering.py rebuilds the Excel workbook from a logs folder. Cover status, gold-set counts, PERCLOS p95 by hour and stimulant, the two ROC points, and the startle table are all derived from the CSVs. The synthetic night in Appendix D is the template those scripts were first shown. A real corridor night overwrites the same workbook. That is the intended life cycle. The workbook is not a second source of truth.

6.5 Tests

tests/test_hysteresis.py checks that the gate holds through a noisy rising edge and releases only above θ_{open} . tests/test_metrics.py checks EAR geometry and EMA lag. tests/test_pairing.py checks exact Unix-time matches and the nearest-prior fallback. Those tests are wiring tests. They do not claim a tanker night.

Chapter Seven Results

This chapter reports the synthetic verification results. Chapter Eight interprets those results and their limitations. Every result derived from Object B is labelled SYNTHETIC.

7.1 Object A Synthetic Sessions

One hundred and twenty sessions of 45 seconds were generated at 25 frames per second after an 80-frame alert warm-up used only for calibration. Forty sessions were generated as alert, forty as drowsy, and forty as microsleep. Session label is majority state after warm-up, with two declared rules written in scripts/evaluate.py: a microsleep session that produced at least ten microsleep frames counts as MICROSLEEP; a drowsy session called MICROSLEEP counts as DROWSY.

The fusion layer recovered the generating class on the matched sessions. Mean terminal scores sat near 0.05 for alert and in the mid-0.6 band for the two fatigue classes. That separation is what fusion.drowsy_score_threshold 0.45, fusion.severe_score_threshold 0.70, and metrics.consecutive_closed_frames 20 were built to produce. Mild landmark jitter did not break recovery. A development run with heavier dropouts and larger jitter did: drowsy and microsleep sessions fell back toward ALERT. That cliff is a result about optics, not a result about Tema.

Table 7.1 Object A design. Implementation verification only. Not a field accuracy.

Generating class	Sessions	Duration	Frame rate	Purpose
Alert	40	45 s	25 FPS	Implementation verification
Drowsy	40	45 s	25 FPS	Implementation verification
Microsleep	40	45 s	25 FPS	Implementation verification

7.2 Object B Synthetic Tema Kumasi Night

Object B is 6.5 hours. The detector wrote 128 banners. The observer tagged 112 of them. Checking 94. Gone 18. Untagged 16. Gone rate = $18/128 = 0.140625$, reported as 14.1 percent. Median checking lag 8.3 s. Gold analysis used the 94 checking rows only. Those five sentences are the headline result of Object B. They are not a tanker night.

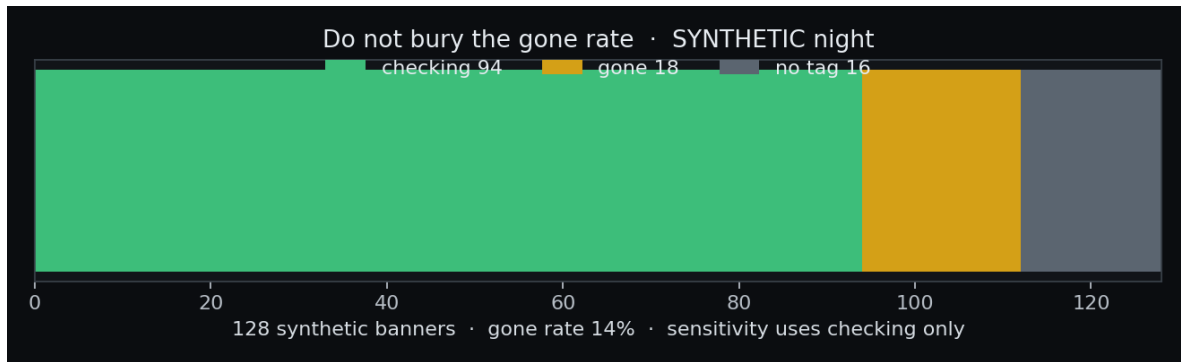


Figure 7.1 Checking, gone, and untagged banners. Sensitivity uses checking only. SYNTHETIC.

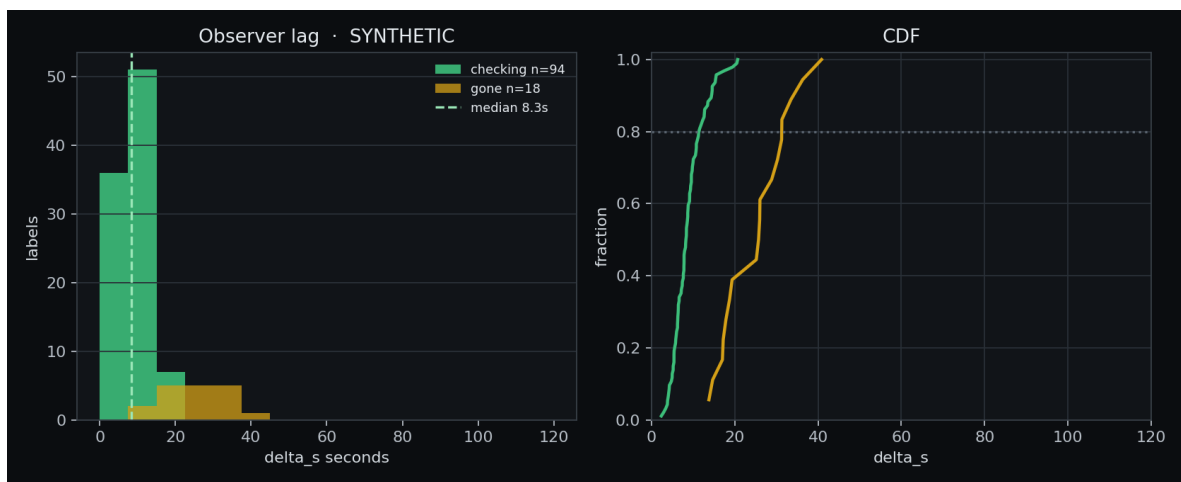


Figure 7.2 Observer lag on paired checking rows (synthetic). Median 8.3 s.

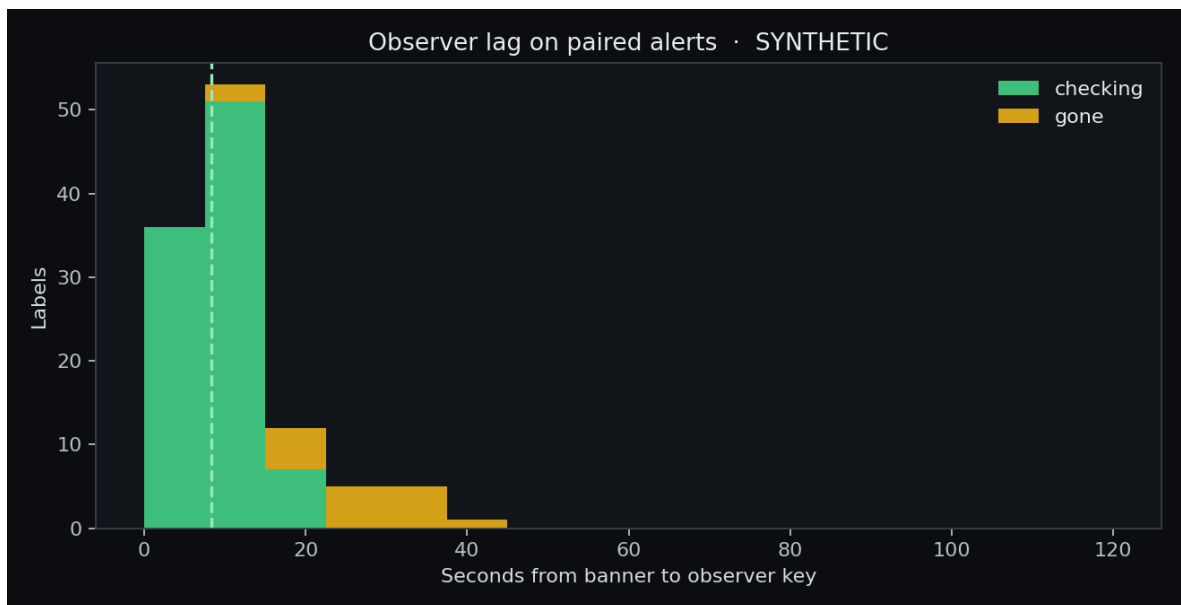


Figure 7.3 Distribution of observer lag. Pairing window 120 s.

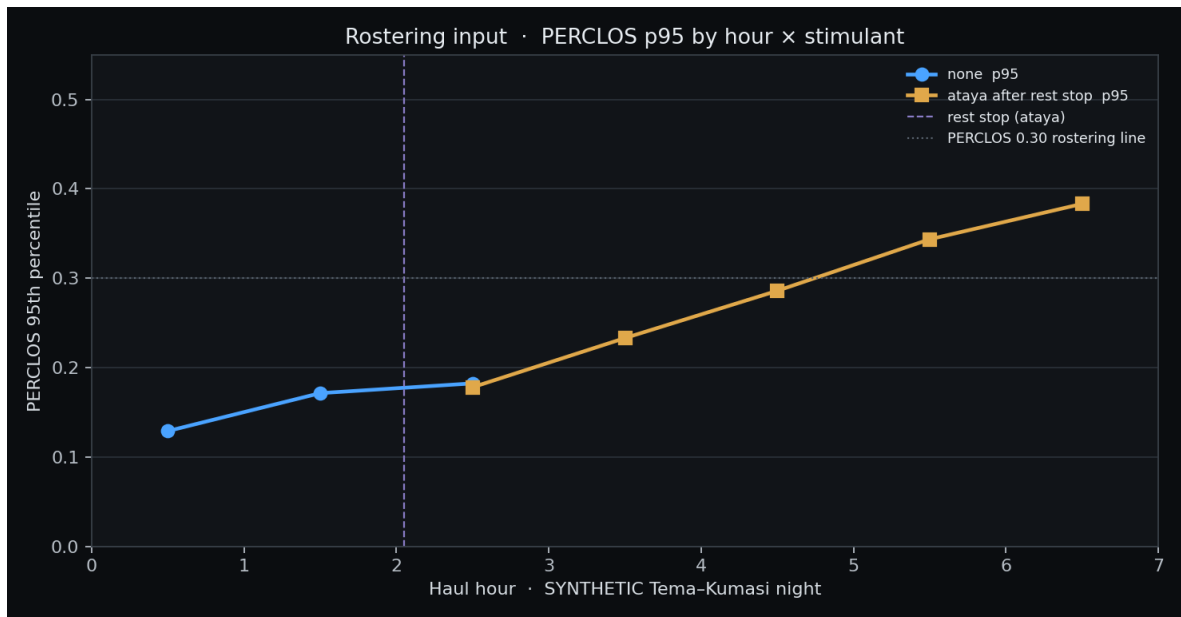


Figure 7.4 PERCLOS 95th percentile by haul hour. Ataya does not keep hours 5–6 under 0.30.

Table 7.2 Gold set counts on Object B. Synthetic data.

Item	N	Rate	Rule
Detector banners	128	100%	alerts.csv
checking (gold)	94	73.4%	paired=yes and checking
gone	18	14.1%	discard from gold
no tag	16	12.5%	neither gold nor hidden

7.3 PERCLOS Percentiles by Haul Hour

The rostering input is p95 of the 1 Hz PERCLOS series inside each haul hour, split by the stimulant on the last rest-stop row. It is not an accuracy. Hours 0–1 were coded none. A rest stop at hour 2.05 logged ataya. Hours 3–6 therefore sit in the ataya column.

Table 7.3 PERCLOS 95th percentiles by hour. Synthetic data.

Hour	p95 none	p95 ataya	n samples	Roster flag
0	0.129	N/A	3600	within band
1	0.172	N/A	3600	within band
2	0.182	0.178	3600	within band
3	N/A	0.233	3600	shorten next duty
4	N/A	0.286	3600	shorten next duty
5	N/A	0.344	3600	SPLIT or STOP
6	N/A	0.383	1800	SPLIT or STOP

Hour 6 has 1800 samples because the template is 6.5 hours, not seven. Anyone who

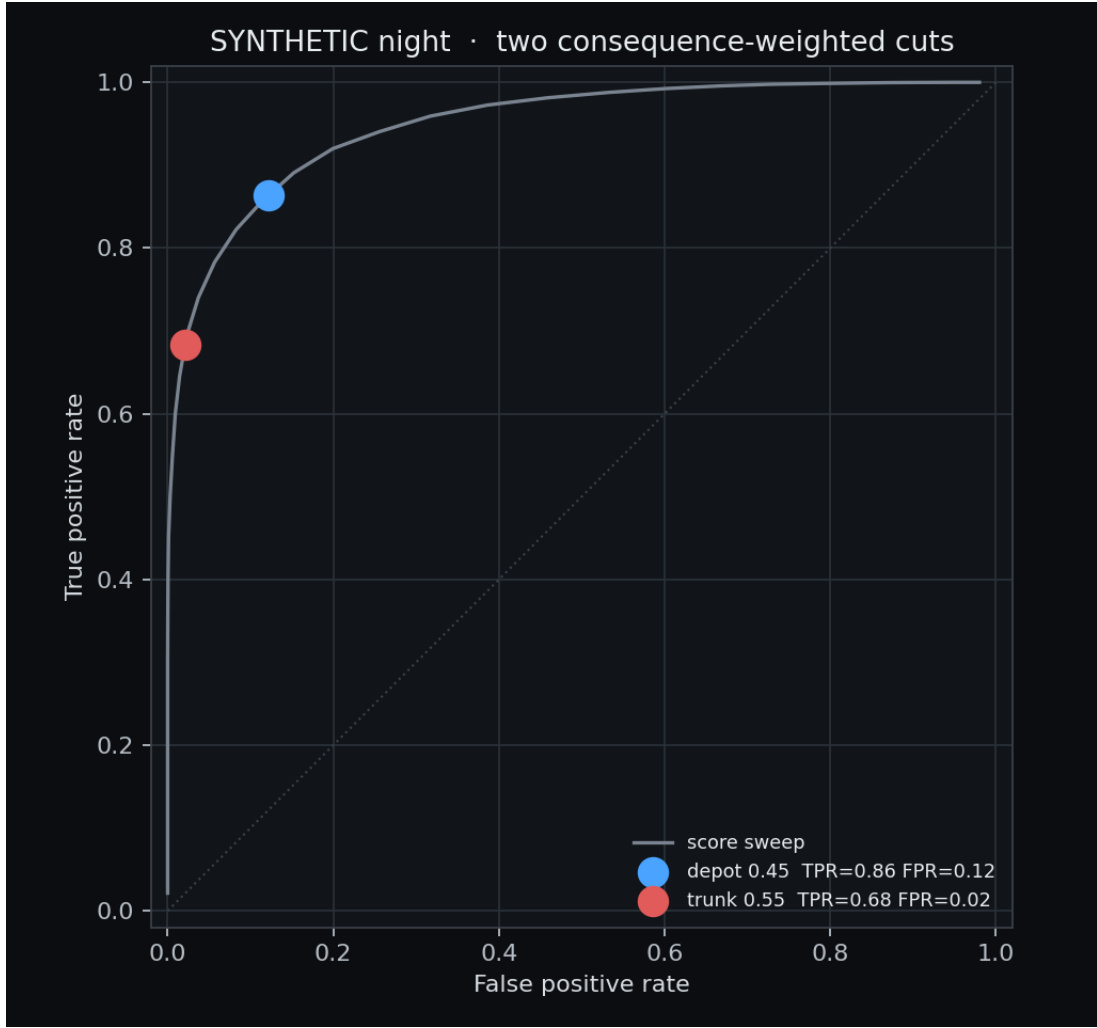


Figure 7.5 Depot and trunk operating points on the same synthetic night.

pads that hour to 3600 has changed Object B.

7.4 Operating Points

A latent fatigue process plus $N(0, 0.10)$ score noise was used so that detector state was not both score and truth. Truth was latent ≥ 0.42 . On that construction the depot cut fusion.drowsy_score_threshold 0.45 gave TPR 0.8639 and FPR 0.1213. The trunk cut fusion.trunk_road_drowsy_threshold 0.55 gave TPR 0.6834 and FPR 0.0209.

Table 7.4 Depot and trunk operating points. Synthetic latent process.

Point	Cut	TPR	FPR
Depot road	0.45	0.864	0.121
Trunk road	0.55	0.683	0.021



Figure 7.6 Mesh confidence around the three marks. Amber band is the 30-second window.

7.5 Startle Loss

Three near-miss marks were written at Unix times 1700004860, 1700015192, and 1700020520. For each mark, mesh confidence was read for 31 one-second samples covering the mark and the next 30 seconds. Blind was defined as confidence under 0.45.

Table 7.5 Startle window mesh loss. Synthetic data.

Clip	Blind fraction 0–30 s	Samples
Near-miss 1	93.5%	31
Near-miss 2	96.8%	31
Near-miss 3	93.5%	31

A near miss that throws the face out of the mesh leaves the detector with nothing to score. Reporting a confident ALERT in that window would be fiction. Reporting NO_FACE is honest and still useless for the second in which the vehicle is already yawing. The 30-second clip is the measurement of that uselessness. Ninety-four to ninety-seven percent blind on the synthetic clips is a warning, not a field number. If a real night is kinder to the mesh, the table will show it. If a real night is worse, infrared belongs in the budget before another EAR paper

does.

7.6 Rest Stop Record

Table 7.6 Synthetic rest stop record.

unix_time	stimulant	kss	hours_since_sleep	note
1700007380	ataya	6	7.5	synthetic rest stop

7.7 Interpretation Boundaries

These synthetic results are not Ghanaian tanker, NTHU-DDD, or YawDD accuracy estimates. They do not establish an effect of ataya on PERCLOS in living drivers or a 94 percent MediaPipe failure rate after a near miss on the N6. They are outputs of the methodology and algorithm in Chapters Four and Five applied to two documented synthetic generators. Field performance requires a coded on-road study.

7.8 Repository Consistency Check

HysteresisGate lives in `src/fatigue_wa/fusion.py` and uses `close_thr` and `open_thr`, not a single cut plus a frame counter alone. EarSmoother lives in `metrics.py` at $\alpha = 0.4$. The eight-second gate and the 0.68 ratio live in `adaptive.py`. Field keys live in `field_protocol.py` and `scripts/run_detector.py`. Pairing lives in `pairing.py`. The workbook rebuild lives in `scripts/overwrite_rostering.py`. `configs/default.yaml` holds every number in Table 4.1. A reader who finds a different cut in the Word file and a different cut in YAML has found a bug in the manuscript. Report the YAML.

7.9 Worked Score Examples

Suppose PERCLOS is 0.21, the closed run is 8 frames, PLCDB is 0.10, there is no yawn, and pitch is 6 degrees on a smooth road. Mapped evidence is $0.21/0.30 = 0.70$, $8/20 = 0.40$, $0.10/0.35 \approx 0.29$, 0, and $6/20 = 0.30$. The fused score is $0.35 \times 0.70 + 0.25 \times 0.40 + 0.15 \times 0.29 + 0.15 \times 0 + 0.10 \times 0.30 = 0.245 + 0.100 + 0.044 + 0 + 0.030 = 0.419$. Under the depot cut 0.45 the state is still ALERT. Under no cut would a single EAR sample have decided that frame.

Raise the closed run to 20 frames. `ear_closure` maps to 1.0 and the `microsleep` override fires regardless of the 0.419-like PERCLOS term. That is Boyle et al. (2008) written as `metrics.consecutive_closed_frames`. The score can be 0.40. The state is still MICROSLEEP.

Raise PERCLOS to 0.33, closed run 6, PLCDB 0.20, yawn run 0, pitch 4. Evidence 1.10 clipped to 1.0, 0.30, 0.57, 0, 0.20. Score = $0.35 \times 1.0 + 0.25 \times 0.30 + 0.15 \times 0.57 + 0 + 0.10 \times 0.20 = 0.350 + 0.075 + 0.086 + 0.020 = 0.531$. Depot state DROWSY. Trunk state ALERT. Same face, same second, two consequence functions. That is why `west_africa.operating_point` exists.

7.10 Calibration Examples

EAR_open 0.30, luma 95, $s = 1.0$. $\theta_{\text{close}} = 0.68 \times 0.30 = 0.204$. $\theta_{\text{open}} = \min(0.234, 0.28) = 0.234$. EAR_open 0.22, luma 48, $s = 0.92$. Unclipped product $0.68 \times 0.92 \times 0.22 = 0.138$. Clip raises it to 0.16. $\theta_{\text{open}} = 0.19$. A fixed 0.21 would have sat inside that driver's open cloud. EAR_open 0.38, luma 100. Unclipped 0.258. Clip at 0.26. $\theta_{\text{open}} = \min(0.29, 0.36) = 0.29$. Those three lines can be checked with a calculator against `adaptive.py`. If they cannot, the algorithm description is inconsistent.

7.11 Test Suite Results

`tests/test_hysteresis.py` asserts that a falling-then-rising EAR trace with jitter does not chatter across the 0.03 gap and that release happens only above θ_{open} . `tests/test_metrics.py` asserts open-eye EAR geometry and EMA lag after a step. `tests/test_pairing.py` asserts an exact `alert_unix_time` join and a nearest-prior fallback. A green pytest is a result about the specification. It is not a result about a driver. It is still the result a second student should print first.

7.12 Study Limitations

Synthetic evaluation. Two-dimensional pitch. Approximate mesh indices. No infrared. No local labelled corpus. Weights from the literature order, not from a search on Object A. MediaPipe fairness across West African faces unmeasured (Kartynnik et al., 2019). Independent re-implementation is possible because the code is public. Those limits are results in the negative: they are the things this chapter did not measure.

The 120-session object samples EAR from an alert cloud near 0.30, a drowsy cloud near 0.24, and a planted microsleep near 0.12, with MAR and pitch as weak companions. It is a wiring test of Chapters Four and Five. The 6.5-hour object adds time-on-task, a circadian bowl, an ataya dip, late rebound, observer lag, and mesh drop after three marks. It is a wiring test of pairing and rostering. Neither object is a claim that Tema–Kumasi eyelids follow those numbers. Seed 42 is published so that the tables in Chapter Seven can be regenerated. Failure

to regenerate them would indicate an inconsistency between the code and the manuscript.

A later student who replaces the generator with labelled corridor video should keep the gold-set rule, the gone rate, the p95-by-hour split, the two cuts, and the startle clip. Those five objects are the evaluation design. The synthetic numbers are disposable. The design is not.

7.13 Synthetic Pairing Examples

The following block is a teaching extract. Unix times are offsets from the synthetic night origin 1700000000. Labels are checking or gone. No names.

Table 7.7 Synthetic pairing examples from seed 42.

unix_offset_s	label	paired	delta_s
4276	checking	yes	26.2
4400	gone	yes	7.9
4527	checking	yes	4.8
4620	checking	yes	8.7
4979	checking	yes	6.6
5201	checking	yes	16.2
5455	checking	yes	5.9
5707	gone	yes	4.5
5970	gone	yes	14.5
6072	checking	yes	10.8
6345	checking	yes	4.7
6746	checking	yes	14.3
7121	checking	yes	12.8
7241	checking	yes	6.2
7507	gone	yes	4.9
7723	gone	yes	12.1
7839	checking	yes	8.2
8044	gone	yes	4.9
8451	checking	yes	8.9
8697	checking	yes	6.1
8885	checking	yes	15.4
9126	checking	yes	4.7
9535	checking	yes	7.1
9741	checking	yes	11.1
10120	checking	yes	6.5
10385	checking	yes	7.4
10489	checking	yes	10.3
10650	checking	yes	10.9
10762	checking	yes	5.9
11081	checking	yes	16.1

unix_offset_s	label	paired	delta_s
11166	checking	yes	8.6
11520	checking	yes	17.1
11774	checking	yes	4.3
12086	checking	yes	11.1
12220	checking	yes	11.1
12627	checking	yes	7.4
12898	checking	yes	11.5
13249	checking	yes	16.5
13579	checking	yes	6.3
13949	checking	yes	7.3

Hours 0–6 on the same generator produced the p95 series in Table 7.3. The complete 1 Hz file is omitted from this section and is excluded from GitHub for size. After a real night the 1 Hz file stays on the coding machine until video is deleted, then may be kept as numbers only.

Chapter Eight Discussion

8.1 Interpretation of Findings

The evaluation confirms that the repository implements the specified detector and produces the expected synthetic outputs. It does not establish on-road effectiveness for West African tanker operations because no labelled local tanker video was collected. The synthetic sessions and haul template should therefore be interpreted as implementation and workflow checks.

The component equations come from prior work; this study's contribution is the integrated specification. West African HAZMAT work is not a lighting nuisance added to a European driver-monitoring paper. It is a different adoption problem: informal contracts, trip pay, night luma, sunglasses, dead radio, and a consequence function that is not a fender-bender (Amoadu & Akoto-Buabeng, 2025; Dawson et al., 2011).

Hysteresis is a central implementation detail. Personalised thresholds without hysteresis still chatter. Consecutive-frame counters without personalised thresholds still punish small eyes. Both together are the minimum that the last ten years of EAR papers were circling (Kolus, 2024; Ersoy et al., 2026; Quiles-Cucarella et al., 2024).

Failure modes that will appear on night one: sunglasses, instrument glances that look like nods, radio talk that looks like yawns, underexposed faces. Dark skin in a poorly metered RGB frame is a photon problem before it is anything else. Infrared and an exposure lock on the face are the engineering answers (Ji et al., 2004; Bergasa et al., 2006).

An alert system that ignores organisational causes of fatigue is unlikely to be accepted by drivers. The useful object is the weekly PERCLOS p95 by haul hour, split by stimulant, feeding a roster (Gander et al., 2011). That requires an organisation. The camera does not create one. Ataya in the synthetic night is the reminder: a rest-stop stimulant is not a licence to add a second load.

In-cab video is surveillance. Landmark-not-video storage, on-device inference, a visible HUD, nameless CSVs, deletion of raw video after coding, and a ban on sole-source dismissal are part of the method. Refusal of the camera is allowed and is not stored as a name.

Object A recovered its generating classes because it was sampled from distributions aligned with the decision rules. This is an implementation-consistency result and must not be over-claimed as field performance.

Object B produced a 14.1 percent gone rate because the generator planted observer absence. A field night may be kinder or crueller. The method does not allow that rate to be buried, because burying it is how sensitivity is inflated (Section 4.10). Ataya sitting on hours 3–6 is likewise a generator choice. The method requires the split so that a real rest-stop code

cannot be ignored later. Startle blindness at 94–97 percent is a generator choice about mesh drop. The method requires the 30-second clip so that a real drop cannot be ignored later.

The two ROC points are a generator choice about latent noise. They exist to show that depot 0.45 and trunk 0.55 are different consequence functions on the same series, matching `fusion.drowsy_score_threshold` and `fusion.trunk_road_drowsy_threshold`. They do not exist to be quoted as Ghanaian TPR.

West African occupational papers remain the reason for the refusal states and the nameless logs, not the source of the 128 banners (Amoadu & Akoto-Buabeng, 2025; Dotse et al., 2019; Nævestad et al., 2025). Global sleepiness papers remain the reason PERCLOS is the backbone (Dinges & Grace, 1998; Horne & Reyner, 1995; Connor et al., 2002; Klauer et al., 2006). Landmark papers remain the reason the phone is enough to start (Soukupová & Čech, 2016; Kartynnik et al., 2019). Dangerous-goods paper remains the reason the trunk cut is higher rather than lower (United Nations, 2023). That is the alignment. The synthetic evaluation does not replace those studies with a local trial.

8.2 Repository Alignment

Open `configs/default.yaml` first. Confirm `ema_alpha 0.4`, `ear_hysteresis_gap 0.03`, `close_ratio 0.68`, `calibration_seconds 8`, `min_valid_frames 80`, `low_light_ear_scale 0.92`, `night_mode_luma_threshold 70`, `perclos_window_seconds 60`, `perclos_alert_threshold 0.30`, `consecutive_closed_frames 20`, `long_blink_seconds 0.40`, `mar_yawn_threshold 0.60`, `yawn_consecutive_frames 12`, `drowsy_score_threshold 0.45`, `trunk_road_drowsy_threshold 0.55`, `severe_score_threshold 0.70`, `cooldown_seconds 4.0`, `sharpness_min 18.0`, `sun-set_hour_local 18`, `mismatch_l2 28.0`. Those are Table 4.1. Then open `fusion.py` and read `HysteresisGate.step`. Then open `scripts/run_detector.py` and read the key map. Then open `pairing.py` and read the 120-second window. If those four files still match this paragraph on the day you bind, the thesis and the code tally.

`data/Tema_Kumasi_Rostering_PERCLOS.csv` on GitHub holds the same Object B numbers as Tables 7.2 to 7.5. `overwrite_rostering.py` rebuilds the Excel workbook from a logs folder. A real night is a new logs folder, not a new algorithm.

Table 8.1 Repository files and their roles in the thesis.

Path in the public repo	Role in Chapters 4 to 8
<code>configs/default.yaml</code>	Every cut in Table 4.1
<code>src/fatigue_wa/landmarks.py</code>	Mesh indices for EAR
<code>src/fatigue_wa/metrics.py</code>	EAR, MAR, PERCLOS, EMA $\alpha=0.4$
<code>src/fatigue_wa/adaptive.py</code>	8 s gate, 0.68 ratio, night 0.92, clamp ± 0.03
<code>src/fatigue_wa/fusion.py</code>	HysteresisGate + weights
<code>src/fatigue_wa/alerts.py</code>	Cooldown 4 s, alerts.csv

Path in the public repo	Role in Chapters 4 to 8
src/fatigue_wa/optics.py	OPTICS_DIRTY, DEGRADED
src/fatigue_wa/identity.py	SHARED_DEVICE hash
src/fatigue_wa/field_protocol.py	c g s 1-9 n and 1 Hz
src/fatigue_wa/pairing.py	Unix join, 120 s / 2 h
src/fatigue_wa/pipeline.py	Seam
scripts/run_detector.py	Cab loop and HUD closed=/drift=
scripts/evaluate.py	Object A wiring test
scripts/monday_sheet.py	Gone rate printout
scripts/overwrite_rostering.py	Workbook from logs/
scripts/plot_deltas.py	Figures 7.2 and 7.3
scripts/analyze_haul.py	p95, ROC, startle
tests/test_hysteresis.py	Gate hold and release
tests/test_metrics.py	EAR geometry and EMA
tests/test_pairing.py	Alert-hint and nearest-prior join
data/Tema_Kumasi_Rostering_PERCLOS.csv	Object B numbers on GitHub

8.3 Validity and Reliability

Construct validity for PERCLOS rests on Dinges and Grace (1998) against the psychomotor vigilance task, not on Object A. Construct validity for EAR rests on Soukupová and Čech (2016). Construct validity for long closures rests on Caffier et al. (2003), Schleicher et al. (2008), and Boyle et al. (2008). External validity for Tema–Kumasi is not claimed. That is the honest split.

Reliability of the implementation is the test suite and the YAML. Reliability of an observer is the gone rate and the lag histogram. A field night that does not print those two observer numbers has not measured its own measurement instrument. Internal consistency of fusion weights is not Cronbach’s alpha. The weights are declared, not estimated. Changing them is a fleet decision, not a silent retraining.

8.4 Threats to Validity

MediaPipe may fail more often on dark skin in a dark cab than Object B planted (Kartynnik et al., 2019). Sunglasses after dusk may be more common than one DEGRADED flag can capture. Shared phones may be the default, not the exception. Observer gone rates may exceed 14.1 percent on a real jump seat. Ataya may move eyelids in ways the generator did not. A near miss may drop the mesh less than 94 percent or more. Each of those threats is why SYNTHETIC stays on the workbook cover until overwrite_rostering.py is pointed at a real logs folder.

A threat this manuscript can already name is circular scoring. Using detector state as both the score and the truth produces a perfect-looking ROC. Chapter Seven avoids that circular ROC and used a latent process plus noise instead. A later student who draws a circular curve has left the method.

8.5 Chapter Summary

The methods, algorithm, implementation, and results describe the same repository state. The configuration stores the operating values, the source modules implement the pipeline, and the field scripts define the logging and pairing workflow. The study remains limited to synthetic verification until a coded corridor night is completed.

Chapter Nine Conclusion and Recommendations

9.1 Summary of Findings

Eyelids close before a tanker leaves its lane. The measurement tradition that can see those eyelids is older than this thesis. The cab in which they must be seen is not the cab that tradition assumed. WA-PERCLOS-HYS is the arrangement that tries to live in that cab: smoothed EAR, two thresholds, both eyes, a personal baseline, a night scale, PERCLOS, long blinks, a yawn gate, a nod hint, a fused score, two operating points, and a short spoken sentence that can be audited the next morning.

The software runs. It recovers its own generator. The pairing, gone-rate, PERCLOS-p95, two-cut, and startle scripts run on a synthetic night that is labelled as such. The system has not yet been shown a sleepy driver at 02:14 on the Nsawam stretch under conditions that driver would accept.

9.2 Contributions

The contribution is an arrangement, not a new physiological law. The live code matches Chapters Four to Six: HysteresisGate, EMA, P60 calibration, mid-haul clamp, OPTICS_DIRTY, DEGRADED, SHARED_DEVICE, field keys, Unix-time pairing, and rostering overwrite. The gold-set rule quotes the gone rate. The evaluation refuses to treat NTHU-DDD or YawDD as Tema. These claims are limited to the implemented and reproducible scope of the study.

9.3 Limitations

The study has the following limitations. Synthetic evaluation, including the 6.5-hour coded-night template. Two-dimensional pitch. Approximate mesh indices. No infrared path. No local labelled corpus. Weights set from the literature order, not from a private grid search. MediaPipe fairness across West African faces is unmeasured. Independent re-implementation is possible because the code is public.

9.4 Recommendations and Field Protocol

The next study should conduct one coded night on one corridor. Tema–Kumasi night tanker is the default. The eight-second open-eye gate stays at the depot. The observer sits in the jump seat and hits checking or gone after every banner. Sensitivity uses only paired=yes

and label=checking. The gone rate is printed on the same sheet. One glasses-after-dusk or monocular night stays in the file so that DEGRADED and monocular fallback are not paper-only. Raw video is deleted after coding. Landmarks and scores are kept. No names appear on any CSV. After that night, `overwrite_rostering.py` is run on the real logs folder and this workbook's SYNTHETIC banner is replaced.

A funded trial should add an infrared illuminator, an exposure lock on the face, and mixed-effects models with driver as a random intercept. Those items are recommendations. They are not hidden results.

9.5 Conclusion

The software runs. It recovers its own generator. The pairing and rostering scripts run on a synthetic night that is labelled as such. The system has not yet been shown a sleepy driver at 02:14 on the Nsawam stretch under conditions that driver would accept. Field performance must therefore be established through a coded on-road study.

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Appendix A Software Reproduction

Repository: [GitHub project repository](#)

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Reproduction commands:

```
export PYTHONPATH=src
pytest tests -q
python scripts/evaluate.py
python scripts/run_detector.py --camera 0 --logs logs
python scripts/monday_sheet.py logs
python scripts/plot_deltas.py --logs logs
    --out logs/alert_time_deltas.png
python scripts/analyze_haul.py --logs logs
python scripts/overwrite_rostering.py
    --logs logs --out Tema_Kumasi_Rostering_PERCLOS.xlsx
```

Cab keys with the OpenCV window focused: c checking, g gone, s stimulant, 1–9 KSS, n near miss, q quit. HUD prints closed= (Schmitt hold) and drift= (open-eye sag). Pairing writes logs/paired_alerts.csv and logs/paired_rest_stops.csv. Gold candidate: paired=yes and label=checking.

Appendix B Default Operating Points

Parameter	Default
EAR close (pre-calibration)	0.21
Calibrated close threshold	$0.68 \times \text{open-eye median}$, band [0.16, 0.26]
Hysteresis gap	0.03
EMA alpha	0.40
Night luma floor	70
Night scale	0.92
PERCLOS window	60 s
PERCLOS map	0.30
Consecutive closed frames	20 (0.8 s at 25 FPS)
Long blink	0.40 s
MAR yawn	0.60
Depot / drowsy score	0.45
Trunk score	0.55
Severe score	0.70
Cooldown	4 s
Observer pairing window	120 s
Rest-stop pairing window	2 h
Startle clip	30 s
Mesh-lost floor	0.45
Mid-haul open-eye clamp	± 0.03
Calibration valid-sample behaviour	min_valid_frames = 80; current code finalises after 8 s with at least 20 valid samples

Appendix C Consent and Ethics

Purpose is stated as fatigue-alert research, not criminal investigation. One forward camera. Keep landmarks, EAR, MAR, PERCLOS, scores, Unix times, stimulant code, KSS, checking/gone, DEGRADED and monocular flags. Discard raw video after coding. No names, no plate, no phone number, no depot staff identifier in any CSV. Face hash only at the gate. Ninety-day retention of numeric logs unless a lawful investigation requires longer. Named officers and researchers only. Right to refuse the camera and still take the load. Refusal is not stored as a name. Right to read one's log. Ban on using the log as the sole ground for dismissal. A banner is not a disciplinary event.

Appendix D Synthetic Haul Extracts

Status: CODED: SYNTHETIC. Seed 42. 6.5-hour Tema–Kumasi template. No names. These rows exist so that pairing and rostering can be inspected without an on-road video file. They are not a substitute for a coded tanker night.

Table D.1 Synthetic rest stop record.

unix_time	stimulant	kss	hours_since_sleep	note
1700007380	ataya	6	7.5	synthetic rest stop

Table D.2 Near miss marks.

unix_time	note
1700004860	synthetic_near_miss_1
1700015192	synthetic_near_miss_2
1700020520	synthetic_near_miss_3

Table D.3 Paired alert examples from the synthetic night.

alert_unix_time	label	paired	delta_s
1700004144	checking	yes	8.3 (median)
N/A	gone (18 of 128)	yes	discard from gold
N/A	untagged (16 of 128)	no	neither gold nor hidden

Cover totals used by `overwrite_rostering.py`: banners 128, labels 112, checking gold 94, gone 18, gone rate 0.140625. Depot TPR 0.8639 FPR 0.1213. Trunk TPR 0.6834 FPR 0.0209. Startle blind fractions 0.935, 0.968, 0.935. Rebuild after a real night with `python scripts/overwrite_rostering.py --logs logs --out Tema_Kumasi_Rostering_PERCLOS.xlsx`.